

DISTINCT SUBSPACES

The “Quotient Space”

NUMERATOR

All Ordered Independent k -tuples in $\mathbb{F}_q^n$

Total size: $(q^n - 1) \cdots (q^n - q^{k-1})$

DENOMINATOR

Size of ONE group:

Number of bases
for a single space

$(q^k - 1) \cdots (q^k - q^{k-1})$

Group 1: Span = W_1
 $(v_1, \dots, v_k)$
 $(v_1 + v_2, \dots, v_k)$
(all bases for W_1)

Forms one
unique space

Subspace W_1

(a distinct k -dimensional plane)

Group 2: Span = W_2
 $(u_1, \dots, u_k)$
 $(u_2, u_1, \dots, u_k)$
(all bases for W_2)

Forms one
unique space

Subspace W_2

(a distinct k -dimensional plane)

$\vdots$

Group m : Span = W_m
 $(w_1, \dots, w_k)$
 $\dots$
(all bases for W_m)

Forms one
unique space

Subspace W_m

(a distinct k -dimensional plane)

$\vdots$

$$\text{Total Distinct Subspaces} = \frac{\text{Numerator (Total } k\text{-tuples)}}{\text{Denominator (Size of one overcount group)}}$$

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_q = \frac{(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1})}{(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1})}$$